Introduction Lecture 1

What is Numerical Analysis?

Study of algorithms that use numerical approximation (as opposed to symbolic manipulations) for problems of mathematical analysis (as distinguished from discrete mathematics)

Find approximate solutions to problems rather than exact ones & obtain error bounds.

Why?

- Sometimes finding an exact solution is impossible.
- Finding an exact solution may take too long.

Our goals: 1) Derive efficient algorithms
2) Estimate their errors

Applications in real life:

- Transportation:
  - Designing fuel-efficient and safe planes, cars, etc., e.g. simulate flow of air around air craft & reduce the need for physical prototypes (more expensive)
    → Simulation of PDEs (partial differential equations)
  - simulate traffic flow

- Weather forecasting:
  - Mathematical models for air motion, pressure, temperature that require numerical solution
  - Simulate hurricanes, tornados, flooding
- Disaster preparedness:
  - Simulate tsunamis
  - Dispersion of pollutants in air, water, soil

- Finance:
  - Option pricing & risk management (also requires probability)
  - Predict economic trends

- Medicine:
  - Image Processing (CTs & MRI)
  - Simulate drug interaction at molecular level

- AI:
  - Train neural nets, estimate generalization errors
  - Robotics: Motion planning & control, testing

- Computer Graphics: Animated movies, games

- Astrophysics:
  - Simulate life of stars
  - Space exploration, rockets, (NASA, SpaceX, ...)

- Sports: Statistics: How likely is a player to score, etc.

1.1. Review of important concepts from Calculus

Limits and Continuity

Definition: A function f defined on a set X of real numbers has the limit L at x_0 written $\lim_{x \to x_0} f(x) = L$ if given any real number $\varepsilon > 0$ there exists $\delta > 0$ such that

$$|f(x) - L| < \varepsilon \quad \text{whenever } x \in X \ \& \ 0 < |x - x_0| < \delta$$

Definition: Let f be a function defined on a set X and let $x_0 \in X$. Then f is continuous at x_0 if

$$\lim_{x \to x_0} f(x) = f(x_0)$$

The function f is continuous on the set X if it is continuous at each number in X.

Limits of sequences

Definition: Let $\{x_n\}_{n=1}^{\infty}$ be an infinite sequence of real/complex numbers. Then the sequence $\{x_n\}_{n=1}^{\infty}$ has a limit x if for any $\varepsilon > 0$ there exists a positive integer $N = N(\varepsilon)$ such that

$$|x - x_n| < \varepsilon \quad \text{whenever} \quad n > N(\varepsilon).$$

The notation $\lim_{n \to \infty} x_n = x$ or $x_n \to x$ as $n \to \infty$

$$x_n \overset{n \to \infty}{\longrightarrow} x$$

means that the sequence $\{x_n\}_{n=1}^{\infty}$ converges to x.

Thm 1.4 If f is a function defined on a set X & $x_0 \in X$, then the following two statements are equivalent:

(i) f is continuous at x_0

(ii) If $\{x_n\}_{n=1}^{\infty} \subset X$ converges to x_0, then

$$\lim_{n\to\infty} f(x_n) = f(x_0)$$

Notation: $C^0([a,b]) = C([a,b]) = \{\text{continuous functions on } [a,b]\}$

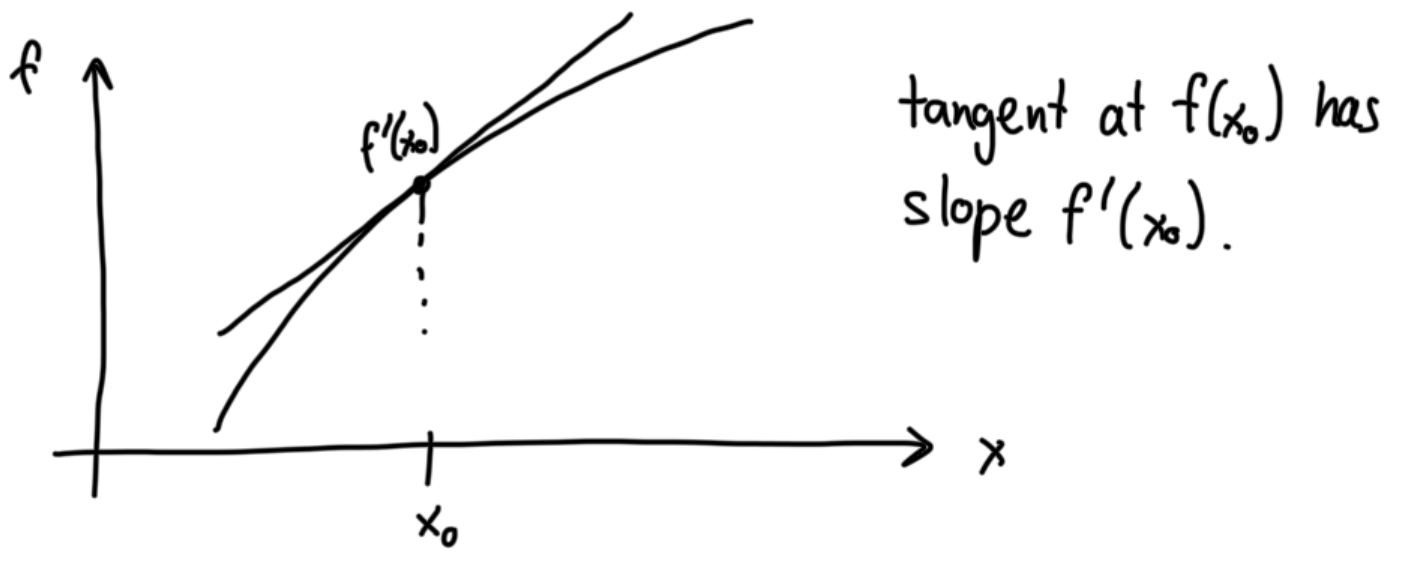
Derivatives:

Definition: Let f be a function defined on an interval (a,b) containing x_0 $x_0 \in (a,b)$. Then f is differentiable at x_0 if

$$f'(x_0) = \lim_{x\to x_0} \frac{f(x)-f(x_0)}{x-x_0} \qquad (1)$$

exists. $f'(x_0)$ is called the derivative of f at x_0

A function that has a derivative at each number in (a,b) is called differentiable on (a,b).



tangent at $f(x_0)$ has slope $f'(x_0)$.